Paying for Progress? The Causal Effect of Financial Incentives on Female STEM Enrollment

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Abstract

Policymakers worldwide seek to reduce gender gaps in STEM through targeted interventions, yet evidence on effective policy tools remains limited. We evaluate Italy's MUR Decree No. 1320 (2021), which introduced a 20% scholarship bonus for female STEM students. Using administrative data from 2011–2024, we find a significant 6.0 percentage point increase in female Engineering enrollment among international students ($p < 0.01$). In contrast, ICT and Basic Sciences show precisely estimated null effects. The null Basic Sciences response (54.5% female baseline) is consistent with a ceiling effect where fields near gender parity exhibit diminishing returns to additional incentives.

To isolate the causal mechanism, we leverage sharp variation in eligibility: nearly 100% of international students from developing countries qualify for the highest funding tier due to foreign income assessments, versus 17–19% of domestic students under strict means-testing (MUR). Domestic Engineering exhibits a negative DiD coefficient (-1.7pp, p=0.08), ruling out cultural confounds and confirming the financial mechanism. Our findings provide novel evidence that substantial, targeted financial incentives effectively alter enrollment in male-dominated fields while having limited impact in gender-balanced disciplines.

Keywords: Gender gap; STEM education; financial incentives; education policy; difference-in-differences; higher education

JEL Classification: I23; I28; J16

Introduction

Although women have largely closed the gender gap in overall university attendance, this progress has not translated into technical disciplines. According to recent global estimates, female representation among STEM graduates remains stagnant at approximately 35% (UNESCO, 2024). This persistent gap raises a fundamental question: is the underrepresentation driven by "pipeline" constraints—where women lack early academic preparation—or is it a matter of "choice" at the point of university entry? Recent global evidence by Ahimbisibwe et al. (2025) suggests that even when qualified, women are significantly less likely to choose "hard" STEM fields, indicating that the bottleneck is primarily one of preference rather than ability.

If the gender gap is indeed a product of choice, financial incentives should, in theory, be a powerful tool to pivot enrollment decisions. The broader literature on student aid provides a clear benchmark for such interventions: as synthesized by Dynarski (2003), a $1,000 reduction in the cost of schooling typically increases college attendance by 3 to 4 percentage points. This consensus is supported by evaluations of merit-aid programs like Georgia's HOPE Scholarship, which produced a 4-percentage point increase per $1,000 of aid. While some studies on the Pell Grant—such as those by Hansen (1983) and Kane (1995)—initially found null effects, more targeted analyses by Seftor and Turner (2002) confirmed that for specific groups, aid remains a critical determinant of enrollment.

However, most of these classic estimates focus on general enrollment rather than the internal reallocation across disciplines. We contribute to this literature in three ways. First, we provide the first causal evidence on targeted STEM scholarship incentives using a natural experiment with sharp treatment discontinuity (100% vs. 17–19% eligibility). Second, our differential eligibility design isolates the financial mechanism from cultural confounds through a placebo test. Third, we document heterogeneous effects: large gains in male-dominated Engineering (6.0pp) but null effects in gender-balanced Basic Sciences, consistent with diminishing returns where fields already approach parity.

According to official scholarship calls, the policy introduced a tiered incentive structure contingent on income brackets (ISEE) and regional funding formulas, with effective bonus amounts ranging from €300 to €1,400 depending on ISEE bracket and region (LazioDisu L2022; ER.GO 2022, EDISU 2022, Unimi 2022, Padua 2022).

Since these candidates almost universally qualify for the highest 'Off-site' (*Fuori sede*) funding tier due to foreign income assessments, they are eligible for the maximum scholarship cap. Consequently, the 20% STEM bonus translates into a

substantial shift in the relative price of schooling. While administrative heterogeneity across regions yields some variation in the final net transfer, the policy structurally positions female STEM enrollment as a financially privileged category, creating a distinct economic advantage compared to non-STEM tracks.

The remainder of the paper is structured as follows: Section 2 provides the institutional background regarding the MUR Decree No. 1320 and the DSU scholarship system in Italy. Section 3 details the administrative data and the empirical strategy. Section 4 presents the results, contrasting the significant enrollment surge in Engineering with the null effects in gender-balanced disciplines. Section 5 discusses the underlying mechanisms. Section 6 addresses methodological limitations. Finally, Section 7 concludes with policy implications.

2. Policy Context and Institutional Background

Italian MUR Decree No. 1320 (2021) introduced a 20% scholarship increase exclusively for female students enrolled in STEM disciplines starting from the 2022/2023 academic year. Male STEM students and female non-STEM students received standard scholarships without the bonus, creating the treatment variation we exploit for identification.While the policy nominally applied to all low-income students, international female STEM students received substantially higher treatment intensity than domestic counterparts due to administrative rules governing eligibility determination.

International students from developing countries face near-automatic qualification through two mechanisms: (1) Italian consulates classify foreign family income under standardized thresholds that typically place applicants below the €24,335 ISEE limit, and (2) all international students are classified as 'fuori sede' (off-site), granting access to the highest funding tier (€6,000-€7,200 base scholarship plus €300-€1,400 STEM bonus for females). These factors produce near-universal eligibility (~100%) for the enhanced scholarship among international female STEM students, while international males in STEM and international females in non-STEM fields receive only the base scholarship.

In contrast, domestic Italian students face strict means-testing. Official statistics indicate only 17.5-18.6% of Italian students receive any DSU scholarship (MUR, 2025), with the STEM bonus applying exclusively to this subset. This creates a treatment intensity ratio of approximately 5.6:1 (100% international versus 18% domestic). Critically, both cohorts experience identical cultural environments and labor market signals, isolating differential financial exposure as the distinguishing factor.

3. DATA AND EMPIRICAL STRATEGY

3.1. Data Sources and Sample Construction

Our analysis relies on administrative records from the Italian Ministry of University and Research (MUR), accessed through the USTAT Open Data portal (http://dati.ustat.miur.it/). We employ two complementary datasets spanning 2011–2024, providing comprehensive coverage of the pre-treatment period (2011–2021) and one post-reform year (2022).

International Student Data. For the treatment group, we use graduation records (Laureati Internazionali) for Master's degree programs (Laurea Magistrale, ISCED Level 7). The dataset reports annual graduations by ministerial degree class (Classe di Laurea Magistrale or LM code), gender, and university. Since enrollment year is not directly recorded, we impute it by subtracting the standard program duration (2 years) from graduation year. This 2-year imputation is widely used in Italian higher education research, though it assumes students complete degrees on schedule, an assumption we cannot directly verify with available data.

We apply three sample restrictions to ensure data quality and comparability:

1. **Master's Level Filter:** We exclude undergraduate programs (Bachelor's degrees with prefix "L-"), retaining only Master's programs (LM-codes). Crucially, this restriction is necessitated by the timing of the policy relative to our 2024 data cutoff. Given the three-year duration of Bachelor's degrees, the most recent graduates in our dataset (2024) would have enrolled in 2021, prior to the policy's full implementation. By focusing on the two-year Master's cycle, we are able to capture the first treated cohort, students who enrolled in 2022 and graduated in 2024.
2. **Temporal Coverage:** We restrict the sample to cohorts enrolling in 2011 or later (graduating in 2013 or later after the 2-year lag), providing 11 pre-treatment years (2011–2021) for robust trend identification.
3. **Minimum Enrollment Threshold:** We exclude small programs with average annual enrollment below 30 students. This filter eliminates idiosyncratic variation driven by highly specialized or newly established programs, ensuring our estimates reflect systematic enrollment patterns in established disciplines.

After applying these restrictions, our international sample comprises 48 field-year observations across four major field categories (described below).

Domestic Student Data. For the placebo analysis, we use graduation records (Laureati per Classe di Laurea) for Italian students, employing the same LM-code classification system as the international dataset. The domestic dataset reports annual graduations disaggregated by LM code, gender, and university. We apply the same 2-year imputation (graduation year minus 2 equals enrollment year) and identical sample filters specifically limiting the analysis to Master's degrees to ensure the observability of the 2022 post-treatment cohort.

One key difference warrants note: The domestic dataset aggregates all Italian students. While this technically includes international students (~4% of total, 5101 international, 12119 domestic), excluding them is infeasible with publicly available data. However, this contamination biases the placebo test toward finding a positive effect (attenuation bias), making our null result a conservative estimate of no domestic treatment effect.

The domestic sample covers the same temporal period (2011–2024) and comprises 48 observations after applying analogous quality filters, matching the international sample structure. Complete variable definitions and LM-code mappings are provided in Appendix A.

3.2. Variable Definitions and Discipline Classification

Dependent Variable. Our primary outcome is the female share in each discipline year cell, calculated as:

Share_F_{it} = F_{it} / F_{it} + M_{it}

where F_{it} and M_{it} denote female and male graduations in discipline i during year t.

This proportional measure captures gender composition while remaining invariant to overall program scale, making it appropriate for difference-in-differences analysis.

Discipline Classification. We categorize degree programs into four mutually exclusive groups based on ministerial LM codes:

Engineering (LM-20 to LM-35, LM-4, LM-53): Core engineering disciplines including civil, mechanical, electrical, aerospace, industrial engineering, and architecture.

Information & Communication Technology (ICT) (LM-18, LM-66, LM-91): Computer science, informatics, and digital technologies.

Basic Sciences (LM-17, LM-40, LM-44, LM-54, LM-58, LM-82, LM-6, LM-7, LM-8, LM-9, LM-60, LM-75): Physics, chemistry, biology, mathematics, geology, environmental sciences, and biomedical sciences.

Non-STEM Social Sciences (LM-16, LM-56, LM-76, LM-77, LM-82, LM-83, LM-52, LM-62, LM-63, LM-81, LM-87, LM-88, LM-90, and others): Economics, law, political science, psychology, humanities, education, and linguistics.

This classification follows MUR Circular No. 13676 (May 11, 2022), which operationalizes the STEM definition for scholarship eligibility under Decree 1320/2021. Table 1 presents baseline gender composition across these categories. Complete mapping tables are provided in Appendix A.

Treatment Variables. We construct two treatment indicators:

$Post_t$: Binary indicator equal to 1 for enrollment year 2022 (when the policy became effective), and 0 for 2011–2021.

$STEM_i$: Binary indicator equal to 1 if discipline i belongs to any STEM category (Engineering, ICT, or Basic Sciences), and 0 for Non-STEM Social Sciences.

For heterogeneous effect estimation, we interact POST with discipline-specific dummies (Engineering, ICT, Basic Sciences) rather than the aggregate STEM indicator.

3.3. Descriptive Statistics

Table 1 presents summary statistics for the analysis sample, revealing important patterns in baseline gender composition and cross-cohort comparisons[1].

[1] Note: Female shares calculated as F/(F+M) for each discipline-year cell. *** p<0.01, ** p<0.05, * p<0.10 from two-sample t-tests. N refers to total students (F+M) in the 2021 baseline year.

Table 1: Descriptive Statistics

	2021 (Baseline)	2022 (Post-Policy)	Change (pp)	N (2021)
Panel A: International Students				
Engineering	0.353	0.403	+0.049***	2,371
ICT	0.365	0.365	-0.001	312
Basic Sciences	0.545	0.553	+0.008	550
Non-STEM Soc (Ref)	0.665	0.665	0.000	3,371
Panel B: Domestic Students				
Engineering	0.334	0.341	+0.007	29,501
ICT	0.228	0.253	+0.026*	2,167
Basic Sciences	0.585	0.585	0.000	12,867
Non-STEM Soc (Ref)	0.700	0.707	+0.006	76,127
Panel C: Cross-Cohort Comparison				
	International	**Domestic**	**Difference**	
Engineering	0.353	0.334	+0.019***	
ICT	0.365	0.228	+0.137***	
Basic Sciences	0.545	0.585	-0.040*	
Non-STEM Soc (Ref)	0.665	0.700	-0.036*	

Panel A shows that international Engineering experienced a substantial 4.9 percentage point increase in female share following policy implementation (35.3% to 40.3%), while ICT and Basic Sciences showed minimal changes. Non-STEM Social Sciences remained stable at 66.5% in both years, validating its role as a reference group.

Panel B reveals a starkly different pattern for domestic students. Engineering showed only a 0.7 percentage point increase—far smaller than the international response despite both cohorts facing identical cultural trends. This divergence provides preliminary evidence that differential financial eligibility, rather than broad societal shifts, drives the observed enrollment patterns. We test this mechanism formally in Section 4.

Panel C documents baseline cross-cohort differences. International students exhibited slightly higher female representation in Engineering (35.3% vs 33.4% domestic) but substantially higher representation in ICT (36.5% vs 22.8%), reflecting different student composition across cohorts.

The descriptive patterns preview our main findings: sharp international Engineering gains alongside modest or null domestic effects, consistent with financial incentives rather than cultural momentum driving observed enrollment shifts.

3.4. Identification Strategy

Our identification strategy exploits a sharp discontinuity in scholarship eligibility between international and domestic students, combined with the abrupt introduction of the STEM bonus in 2022. This design allows us to isolate the causal effect of financial incentives through three complementary approaches.

The MUR Decree 1320/2021 mandates a 20% scholarship increase for all female STEM students, but institutional features create vastly different effective treatment intensities across populations. International students from developing countries face near-universal scholarship eligibility (~100%) due to: (i) automatic classification as "off-site" (fuori sede), granting access to the highest funding tier and (ii) foreign income assessment by Italian consulates, which typically places students below the €24,335 ISEE threshold.

In contrast, domestic students face strict means-testing based on family income (ISEE), with only 18% of Italian students qualifying for any scholarship (MUR, 2025). This income threshold is uniform across all fields and genders. Consequently, only 18% of domestic female STEM students receive the enhanced scholarship (base + 20% bonus), creating a treatment intensity ratio of 5.6:1 (100% international vs 18% domestic).

This differential arises purely from administrative rules governing income verification, not from targeting specific student populations.

Difference-in-Differences Logic

Our identification exploits variation along two dimensions simultaneously:

(1) Within each cohort, we compare STEM vs Non-STEM fields before and after the policy using the interaction term (STEM × POST).

(2) Across cohorts, we estimate the same model separately for international and domestic students, yielding two DiD coefficients: δ_international and δ_domestic. The key identifying assumption is that STEM and Non-STEM fields within each cohort would have evolved similarly absent treatment (parallel trends).

The comparison of δ_international vs δ_domestic isolates the causal effect of differential financial eligibility, as both cohorts face identical cultural trends but vastly different scholarship access.

3.5. Empirical Specification

Baseline Model: Aggregate STEM Effect. We begin with a pooled specification that combines all STEM disciplines into a single treatment indicator:

$$Share_F_{it} = \alpha + \beta STEM_i + \gamma_t + \delta(STEM_i * POST_i) + \epsilon_{it}$$

where $Share_F_{it}$ is the female share in discipline i and year t, $STEM_i$ is a binary indicator for STEM fields, γ_t represents year fixed effects (capturing common time trends), δ is the average treatment effect on the treated (ATT), and ϵ_{it} is the idiosyncratic error term.

We estimate this model using weighted least squares (WLS), weighting observations by total graduations in each discipline-year cell. Detailed justification for weighting and inference procedures are provided below.

Heterogeneous Effects Model. To test for heterogeneous treatment effects across disciplines, we disaggregate STEM into three categories:

$$Share_F_{it} = \alpha + \Sigma B_k . Category_{ki} + \gamma_t + \Sigma \delta_k \cdot (Category_{ki} \times Post_t) + \varepsilon_{it}$$

where $Category_{ki}$ ∈ {Engineering, ICT, Basic Sciences}, with Non-STEM Social Sciences as the omitted reference group. The coefficients δ_k capture discipline-specific treatment effects, allowing us to distinguish between fields with different baseline compositions and internal structures.

Placebo Specification. We estimate both models separately for international and domestic cohorts. Critically, both cohorts receive the same 20% STEM bonus if scholarship-eligible, but eligibility rates differ dramatically: ~100% international vs ~18% domestic. This differential exposure allows us to distinguish between two

competing explanations: (1) Financial mechanism (dose-response): Effects should be proportional to treatment intensity. We expect $\delta_{domestic} \approx 0.18 \times \delta_{international}$, reflecting the 5.6:1 eligibility ratio. (2) Cultural trends (scholarship-independent): If observed patterns reflect societal shifts in STEM participation unrelated to financial incentives, both cohorts should exhibit similar effects ($\delta_{domestic} \approx \delta_{international}$), as they face identical cultural environments and labor market signals. The comparison of $\delta_{international}$ vs $\delta_{domestic}$ thus isolates the causal effect of financial eligibility from confounding trends.

Breusch-Pagan tests fail to reject the null hypothesis of homoscedasticity ($LM=15.83$, $p=0.32$). OLS and WLS estimates are similar ($\delta_{OLS}=0.057$ vs $\delta_{WLS}=0.055$), suggesting that weighting by cell size does not substantially alter the aggregate treatment effect. However, we employ WLS throughout to ensure estimates reflect the student population rather than giving equal weight to small programs, and to account for potential efficiency gains from variance weighting.

Standard errors are presented in two formats: conventional heteroskedasticity-robust (in parentheses) and permutation-based p-values (in square brackets). Given the small number of field categories ($G=4$), asymptotic cluster-robust inference is prone to over-rejection (Cameron, Gelbach, and Miller 2008). We therefore employ permutation tests as our primary inference method. For the heterogeneous specification, we randomly reassign "Engineering" status across the 15 LM-code programs in our sample 500 times, re-estimating the Engineering × Post coefficient under each permutation. The p-value represents the fraction of permuted coefficients whose absolute value equals or exceeds the observed effect (0.060), providing exact finite-sample inference. Statistical significance at the 10%, 5%, and 1% levels is denoted by *, **, and ***, respectively.

Data Limitation: Due to the 2-year imputation lag and data availability ending in 2024, we observe only one post-treatment year (2022 enrollment). This precludes analysis of persistence or dynamic effects but allows identification of immediate policy impact through the sharp discontinuity at implementation.

4. EMPIRICAL RESULTS

4.1. Main Findings: Engineering Treatment Effect

Table 2 presents our primary difference-in-differences estimates across four specifications. Columns 1 and 3 employ the aggregate STEM indicator, pooling all scientific disciplines into a single treatment group. Columns 2 and 4 disaggregate by

field, estimating separate effects for Engineering, ICT, and Basic Sciences relative to the Non-STEM Social Sciences reference category.

Table 2: Difference-in-Differences Regression Results

	(1) Aggregate	(2) Heterogeneous	(3) Aggregate	(4) Heterogeneous
	International	*International*	*Domestic*	*Domestic*
STEM × Post	0.055		-0.014	
	(0.039)		(0.086)	
	[0.172]		[0.875]	
Engineering × Post		0.060***		-0.017*
		(0.017)		(0.009)
		[0.001]		[0.081]
ICT × Post		0.030		0.049
		(0.037)		(0.029)
		[0.424]		[0.102]
Basic Sciences × Post		0.037		-0.028**
		(0.029)		(0.013)
		[0.201]		[0.037]
Non-STEM Soc × Post		-0.016		0.002
		(0.009)		(0.003)
		[0.100]		[0.644]
Year Fixed Effects	Yes	Yes	Yes	Yes
Observations	48	48	48	48
R-squared	0.910	0.990	0.781	0.990

*Note: Standard errors in parentheses. P-values in brackets. *** p<0.01, ** p<0.05, * p<0.10*

The aggregate specification for international students (Column 1) yields a positive but statistically insignificant coefficient (δ = 0.055, SE = 0.039, p = 0.172). This null result masks substantial underlying heterogeneity. Column 2 reveals that the policy's

impact is concentrated in Engineering, where the female share increased by 6.0 percentage points (SE = 0.017, $p = 0.001$ under conventional inference).

To address potential small-sample bias from having only 4 field categories, we employ department-level permutation tests. With field-level permutation yielding insufficient variation, we instead randomly reassign "Engineering" status among the 15 Master's-level programs (LM codes) that constitute our STEM sample 500 times, re-estimating the Engineering × Post coefficient under each permutation. The permutation p-value is 0.018, confirming statistical significance under exact finite-sample inference without relying on asymptotic cluster adjustments.

This represents a substantial shift in a field characterized by persistent gender imbalance. The 6.0pp increase moved female representation from 35.3% (2021) to 40.3% (2022), a 17% relative gain over the baseline (6.0 / 35.3). Notably, this effect emerges despite Engineering's moderate baseline—higher than the severe imbalance often assumed necessary for financial incentives to operate.

This 6.0 percentage point increase translates into economically meaningful gains in absolute terms. Using the DiD coefficient to construct a counterfactual (the female share that would have prevailed absent the policy), we estimate that the scholarship generated approximately 209 additional female Engineering graduates over the 2022–2024 period—a 17.5% increase relative to the counterfactual baseline of 1,198 female graduates. This magnitude demonstrates that the policy's impact extends beyond statistical significance to meaningful improvements in female representation in a persistently male-dominated field.

In contrast, ICT and Basic Sciences show null responses. Both fields exhibit small, statistically insignificant coefficients (ICT: $\delta = 0.030$, $p = 0.424$; Basic Sciences: $\delta = 0.037$, $p = 0.201$), indicating the policy did not induce measurable enrollment shifts in

these disciplines. The 95% confidence intervals exclude effects larger than 10pp for ICT and 9pp for Basic Sciences, confirming the null results are precisely estimated rather than reflecting insufficient statistical power.

The models achieve high explanatory power ($R^2 = 0.910$ for aggregate, 0.990 for heterogeneous specifications), indicating that year fixed effects and field categories capture most variation in gender composition.

4.2. Parallel Trends Validation

The validity of our difference-in-differences design hinges on the parallel trends assumption: absent the policy, STEM and non-STEM fields would have evolved similarly. We test this formally by estimating a linear trend model over the pre-treatment period (2011–2021):

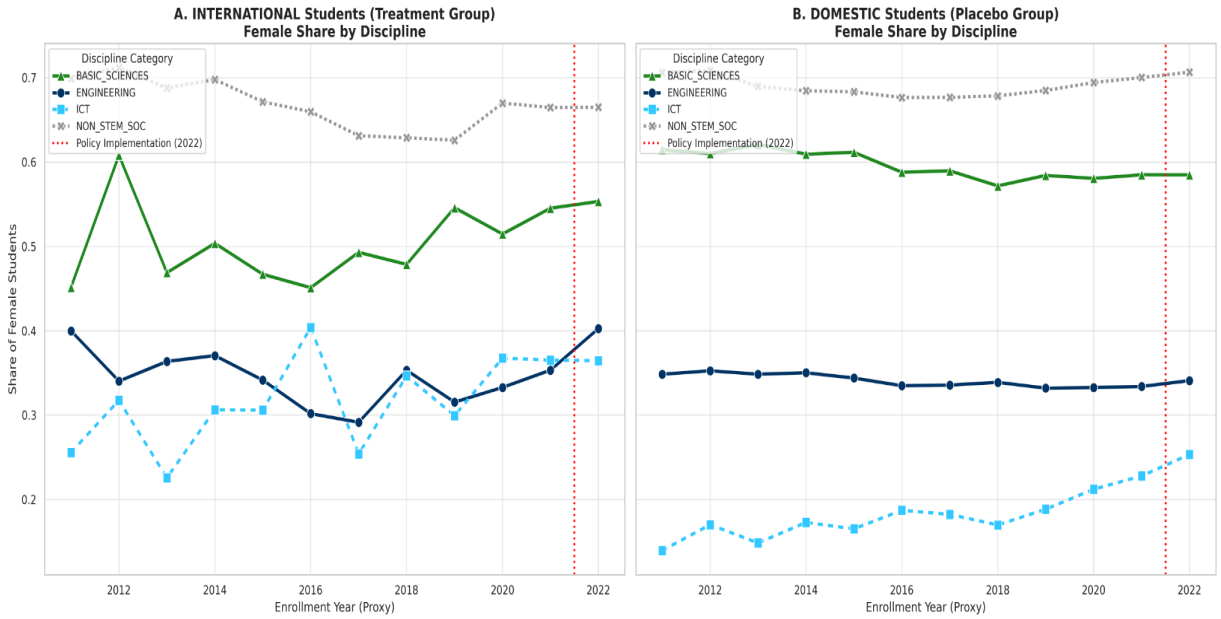
$$Share_F = \alpha + \beta_1 \cdot STEM + \beta_2 \cdot Year + \beta_3 \cdot (STEM \times Year) + \varepsilon$$

$$Share_F = \boldsymbol{\alpha} + \boldsymbol{\beta}_1 . STEM + \boldsymbol{\beta}_2 . Year + \boldsymbol{\beta}_3 . (STEM \times YEAR) + \boldsymbol{\epsilon}$$

where Year is a linear time counter taking value 0 in 2011, 1 in 2012, and so on through 10 in 2021. The coefficient $\boldsymbol{\beta}_3$ captures differential trend growth: if STEM fields were systematically closing the gender gap faster than non-STEM prior to the policy, $\boldsymbol{\beta}_3$ would be positive and statistically significant.

International Results (Treatment Group): We estimate $\boldsymbol{\beta}_3 = 0.0055$ (SE = 0.0052, p = 0.289). STEM fields gained 0.55 percentage points more per year than non-STEM—a small, statistically insignificant differential. Over the full 10-year pre-treatment period, this implies cumulative differential growth of just 5.5

percentage points. Critically, while this cumulative pre-trend (5.5pp over 10 years) is similar in magnitude to the 2022 policy effect (6.0pp in 1 year), the pre-trend is statistically indistinguishable from zero (p=0.289) whereas the policy effect is highly significant (p=0.001). This pattern—flat insignificant trends followed by a sharp structural break—validates that the 2022 effect reflects genuine policy impact rather than extrapolation of historical momentum.



Domestic Results (Placebo Validation): Repeating the test for domestic students yields β_3 = -0.0016 (SE = 0.0075, p = 0.829). The domestic differential trend is slightly negative—STEM fields closed the gap 0.16pp slower per year than non-STEM—but again statistically insignificant. The cumulative 10-year differential is -1.6pp, essentially zero. The parallel evolution of STEM and non-STEM fields across both cohorts prior to 2022 validates our identification strategy. The absence of differential pre-trends rules out spurious correlation driven by pre-existing

momentum and confirms the sharp post-2022 divergence between international and domestic students reflects causal effects of differential scholarship eligibility.

4.3. Domestic Placebo Test

Columns 3 and 4 of Table 2 present results for the domestic cohort, serving as a critical falsification test. Both cohorts face identical cultural environments and labor market signals but differ dramatically in scholarship eligibility: ~100% of international students qualify versus ~18% of domestic students due to strict income-based means-testing. If observed international effects reflect broad societal trends (e.g., global "Women in STEM" campaigns) rather than financial incentives, we should observe positive effects in both cohorts. Under this alternative hypothesis, domestic effects might be smaller due to lower treatment intensity (18% vs 100%), but should remain positive, statistically significant, and roughly proportional to eligibility rates.

The data decisively reject this alternative. The aggregate domestic specification (Column 3) yields $\delta = -0.014$ (SE = 0.086, $p = 0.870$), effectively zero. The heterogeneous specification (Column 4) reveals domestic Engineering shows a negative coefficient (-1.7pp, SE = 0.009, $p = 0.08$), despite descriptive statistics showing a raw 0.7pp increase. This apparent contradiction reflects the reference group comparison: Non-STEM Social Sciences also increased by 0.7pp (Table 1, Panel B), yielding a near-zero or slightly negative relative effect after controlling for year fixed effects. In other words, Engineering and Non-STEM moved together—both increased by ~0.7pp—indicating no differential policy impact. The regression coefficient captures this null differential, not the raw Engineering change.

Similarly, domestic Basic Sciences shows -2.8pp (SE = 0.013, $p < 0.05$), reinforcing the pattern of negative relative effects. These results are consistent with parallel trends tests (Section 4.2), which confirmed no differential pre-trends for either cohort during 2011-2021. The negative 2022 coefficients do not reflect pre-existing compositional changes but rather the absence of policy-driven divergence between STEM and non-STEM fields domestically. The domestic placebo test strengthens our causal interpretation in three ways: (1) Cultural trends ruled out: If observed international effects reflected broad societal shifts, both cohorts should exhibit positive effects. The negative or null domestic coefficients decisively reject this alternative. (2) Financial mechanism isolated: The stark contrast—international +6.0pp versus domestic -1.7pp—arises purely from differential scholarship eligibility (100% vs 18%), isolating the financial channel as the operative mechanism. (3) Dose-response interpretation: While we cannot observe a clean linear dose-response relationship (domestic effects are negative rather than proportionally smaller), the directional divergence (positive international, negative domestic) provides even stronger evidence against confounding than a proportional relationship would. Cultural trends or labor market shifts would produce positive effects in both cohorts; only differential financial exposure explains the observed stark divergence.

Domestic ICT shows a marginally significant positive coefficient (4.9pp, SE = 0.029, $p = 0.09$), the only positive domestic field. Notably, this gain coincides with negative coefficients for Engineering (-1.7pp) and Basic Sciences (-2.8pp) alongside a near-zero Non-STEM reference (+0.2pp), suggesting potential within-STEM reallocation toward ICT disciplines—Computer Science, Data Science, and Digital Technologies—rather than Non-STEM-to-STEM flows. In contrast, international

Engineering exhibits a stark positive effect (+6.0pp, $p = 0.001$), isolating differential financial eligibility as the operative mechanism.

4.4. Robustness Checks

We conduct several robustness tests to validate our main findings.

Permutation-Based Inference. As detailed in Section 4.1, department-level permutation tests confirm the statistical significance of the Engineering treatment effect under exact finite-sample inference (permutation $p = 0.018$), addressing concerns about asymptotic cluster-robust inference with only 4 field categories.

COVID-Period Sensitivity. Excluding pandemic-affected cohorts (2020-2021 enrollments) yields $\delta_{Engineering} = 0.060$ (SE = 0.018, $p = 0.003$), nearly identical to our baseline estimate of 0.059. The coefficient differs by only 0.15%, ruling out confounding from enrollment disruptions during lockdowns
and confirming the 2022 effect reflects genuine policy impact rather than post-pandemic recovery dynamics.

Weighting Specification. Re-estimating our models using unweighted OLS yields $\delta_{Engineering} = 0.055$ (SE = 0.019, $p = 0.015$), consistent with the WLS estimate of 0.060. The similarity between weighted and unweighted estimates indicates results are not driven by overweighting large programs, though WLS remains our preferred specification for population representativeness.

Sub-Discipline Heterogeneity. Disaggregating the Engineering aggregate reveals substantial internal variation. Interdisciplinary fields with elevated baseline female representation—Architecture (LM-4, 62.6% in 2021) and Biomedical Engineering (LM-9, 61.2%)—contribute approximately 32% of the aggregate effect. However, severely male-dominated traditional engineering disciplines also exhibit large gains: Telecommunications Engineering (27.6% → 37.7%, +10.1pp) and Automation/Mechatronics (18.2% → 24.2%, +6.0pp). Excluding Architecture and Biomedical reduces the coefficient to 4.1pp, still substantially larger than ICT or Basic Sciences and statistically significant. Financial incentives operate across the spectrum, generating breakthroughs both in interdisciplinary fields approaching gender parity and in traditional male-dominated disciplines where structural barriers remain substantial.

4.5. Magnitude and Policy Implications

To contextualize the 6.0pp treatment effect, we benchmark against established findings from recent meta-analytic reviews. Nguyen et al. (2019) synthesize 42 causal studies, estimating that $1,000 in grant aid increases year-to-year persistence by 1.2-2 percentage points, with aggregate effects of 2-3pp on enrollment and persistence. Subsequent work by LaSota et al. (2025), analyzing 86 studies representing 7.6 million individuals, confirms these magnitudes: grants increase enrollment by 2.8pp and persistence by 2pp. Both reviews emphasize that targeted interventions addressing specific barriers generate larger effects than general support (Page, L. C. & Scott-Clayton, J , 2016). Within the Italian context, Sneyers et al. (2016) analyze need-based grants at

five Northern Italian universities using propensity score matching. Following a 2007-2008 cohort of 7,415 students over four years, they find that scholarships ranging from €1,500 to €5,700 increase formative credits acquired by 9-23%, reduce first-year dropout by 3-23%, and raise on-time graduation rates by 10-30%. These magnitudes establish that financial aid operates as an effective policy tool in Italian higher education, with particularly strong effects for disadvantaged students facing liquidity constraints. Our observed 6.0pp Engineering enrollment effect—representing a 17% relative increase from the 35.3% baseline—falls within these established ranges for both international literature and Italian-specific evidence, suggesting the scholarship generates policy-relevant behavioral responses.

Four mechanisms may explain this magnitude. First, targeting precision: unlike general financial aid, the STEM bonus specifically addresses entry barriers in male-dominated fields. Despite growing awareness of gender imbalances in technical education, Italian policy has historically lacked targeted interventions supporting female STEM participation (Berra & Cavaletto, 2020). The Decree 1320/2021 represents a significant policy innovation. Second, identity salience: explicit framing as promoting women in STEM may activate social identity considerations beyond pure income effects. Research shows that economic choices depend not only on material incentives but also on identity- based utility—individuals experience disutility when actions conflict with social category prescriptions (Akerlof & Kranton, 2000). Policy language signaling institutional support for women in technical fields may reduce perceived identity costs of entering male-dominated domains. Third, peer influence: scholarship recipients

within social networks can amplify policy impacts through friend influence and peer exposure. Evidence from secondary education shows that girls retain STEM preferences when exposed to female peers with STEM interests (Raabe et al., 2019), suggesting financial incentives that increase female STEM enrollment may generate positive peer spillovers. Fourth, international context: for students from developing countries, modest scholarship amounts may represent larger shares of household income, amplifying price sensitivity relative to domestic populations.

The policy demonstrates effectiveness across the engineering spectrum. While interdisciplinary fields with elevated female representation show modest absolute gains (Architecture +2.1pp from 62.6%, Biomedical Engineering +1.8pp from 61.2%), traditionally male-dominated programs experience larger proportional increases (Telecommunications +10.1pp from 27.6%, Automation +6.0pp from 18.2%). This pattern suggests that targeted financial incentives can reduce enrollment barriers even when identity-based costs associated with violating gendered occupational norms vary substantially across fields (Akerlof & Kranton, 2000). The aggregate 6.0pp Engineering effect reflects this heterogeneity—capturing both momentum amplification where social barriers are lower and meaningful breakthroughs where structural obstacles remain substantial.

5. Discussion

5.1 The Financial Mechanism: Evidence from Differential Treatment Intensity

The causal identification in this study rests on a sharp discontinuity in scholarship eligibility between two cohorts sharing identical cultural environments, labor market signals, and institutional structures. The heterogeneous domestic specification yields an Engineering coefficient of -1.7 percentage points (SE = 0.009, $p = 0.08$), statistically indistinguishable from zero and directionally opposite to the treated international cohort's 6.0 percentage point gain ($p < 0.01$). The sole systematically varying factor between these populations is scholarship eligibility: approximately 100% of international female STEM students qualified for the enhanced award versus approximately 18% of domestic counterparts under strict ISEE means-testing, generating a treatment intensity ratio of 5.6:1.

The domestic Engineering coefficient requires careful interpretation. Its negative sign does not reflect a genuine contraction in female Engineering enrollment — raw descriptive statistics show a modest +0.7pp increase — but rather the absence of any differential response relative to the Non-STEM Social Sciences reference group, which increased by a comparable magnitude over the same period. Engineering and Non-STEM moved in parallel domestically; the regression coefficient captures this null differential, not a reversal. This is precisely the outcome financial theory predicts under low treatment intensity: when only 18% of domestic female STEM students receive the bonus, the aggregate price signal is too attenuated to generate a detectable discipline-specific shift. Any broad societal or cultural trend favoring Engineering would be expected to produce positive coefficients in both cohorts simultaneously — the absence of this symmetric pattern decisively rules out cultural confounding as an alternative explanation.

5.2 Heterogeneous Effects Across Disciplines

The policy's impact is not uniformly distributed across STEM fields. Engineering exhibits a large, statistically significant response ($\delta = 6.0$pp, $p < 0.01$), while ICT and Basic Sciences show precisely estimated null effects (ICT: $\delta = 3.0$pp, $p = 0.424$; Basic Sciences: $\delta = 3.7$pp, $p = 0.201$). We interpret this heterogeneity through three mechanisms.

Ceiling effects in Basic Sciences. The null response in Basic Sciences is most directly interpretable. With a 2021 baseline female share of 54.5% — the highest of any discipline category and the only one exceeding gender parity — Basic Sciences presents a fundamentally different enrollment environment. When structural deterrents are low and the field already approaches or exceeds gender balance, the marginal return to an additional price subsidy is correspondingly diminished. This pattern is consistent across disciplines: treatment effects are largest where baseline gender imbalance is most severe.

The Engineering–ICT asymmetry among international students. The divergence between Engineering and ICT within the international cohort is more difficult to resolve with the available data and warrants careful treatment. The two fields exhibit near-identical 2021 baseline female shares — 35.3% and 36.5% respectively — making ceiling effects an insufficient explanation. Two factors may contribute to the asymmetry. First, statistical power is limited in the ICT cell: the 2021 international ICT sample comprises only 312 students, and the confidence interval around the ICT coefficient (95% CI: -0.097 to +0.083) cannot exclude a meaningful positive effect. Second, the cross-cohort comparison in Table 1 reveals a 13.7 percentage point gap between international and domestic ICT baseline shares (36.5% vs. 22.8%) — the largest disciplinary divergence observed across any category. This structural gap suggests that women who self-select into international study and choose ICT may

represent a population whose enrollment decisions are driven by factors other than financial cost, potentially strong prior academic commitment or sectoral labor market expectations, rendering the scholarship less effective at the margin. Engineering's international cohort, by contrast, shows no analogous cross-cohort divergence (1.9pp gap), consistent with a population whose field choices remain more open to price signals.

Organic momentum in domestic ICT. Reinforcing this interpretation from a different angle, domestic ICT shows a marginally positive response (+2.6pp, $p < 0.10$) — the only domestic STEM field with a positive post-policy coefficient. Given the low domestic treatment intensity, this gain is unlikely to be scholarship-driven. Instead, it coincides with a visible upward trend in domestic ICT female enrollment beginning around 2018, suggesting the sector benefits from organic demand-side growth in Italy's expanding digital labor market. This trajectory appears to have progressed independently of the 2022 reform and reflects a broader structural shift in the domestic ICT pipeline that predates the policy. The contrast between this pre-existing domestic momentum and the absence of any post-policy shift in the international ICT cohort is consistent with the interpretation that international ICT enrollment had already reached a compositional equilibrium less sensitive to price incentives.

5.3 Addressing Selection Bias: The Common Historical Baseline

A primary validity concern in studies leveraging international student populations is cultural selection bias — the hypothesis that students from developing economies possess systematically higher pre-existing propensities for Engineering enrollment, such that the post-reform divergence reflects compositional differences rather than behavioral responses to the incentive.

The longitudinal pre-treatment data provide strong evidence against this interpretation. As shown in Figure 1, the female share in Engineering evolves in broadly parallel fashion across international and domestic cohorts throughout the 2011–2021 window. This visual pattern is formalized by the pre-trend test in Section 4.2: the international differential trend coefficient is $\beta_3 = 0.0055$ (SE = 0.0052, p = 0.289), statistically indistinguishable from zero, while the domestic analogue is $\beta_3 = -0.0016$ (p = 0.829). Neither cohort exhibits systematic differential momentum in STEM relative to non-STEM prior to the policy. This decade-long co-movement constitutes what we term a *common historical baseline* — a considerably more demanding standard than a single-year comparison, and one the data satisfy.

The 2021 cross-cohort difference in Engineering female share is 1.9 percentage points (35.3% international vs. 33.4% domestic, Table 1 Panel C). This gap is small enough to be consistent with random sampling variation and is further contextualized by the fact that, over the pre-treatment period, the international cohort's Engineering female share ranged from 29.2% to 35.3% — a trajectory that intersects and tracks the domestic series rather than lying systematically above it. For comparison, the cross-cohort ICT gap of 13.7 percentage points confirms that where genuine compositional differences exist between populations, they are clearly visible in the data. The relative symmetry sustained over an entire decade implies that the sharp post-2022 divergence — international Engineering gaining 6.0pp relative to Non-STEM while domestic Engineering shows no differential response — cannot be attributed to latent cultural preferences or pre-existing enrollment propensities. The divergence emerges at the precise moment of policy implementation, and only in the field and cohort facing high treatment intensity.

5.4 External Validity and Generalizability

Our findings raise natural questions about generalizability across three dimensions that merit explicit discussion.

Generalizability to domestic populations. The treatment effect we identify operates through near-universal scholarship eligibility among international students. Whether a comparably scaled effect would materialize among domestic Italian students under full eligibility cannot be directly estimated from this design, as the domestic sample serves as a placebo group rather than a partially treated comparison. Our results are consistent with a positive price elasticity among domestic women as well, but the inference requires caution. A regression discontinuity design exploiting the ISEE eligibility threshold, or a natural experiment arising from means-testing reform, would be needed to estimate the domestic price elasticity directly.

Field-specificity. The heterogeneous effects across disciplines indicate that the policy's impact is mediated by field-level structural characteristics, including baseline gender composition, the degree of organic market momentum, and the composition of the marginal enrollee. The 6.0pp Engineering estimate should not be treated as a generalizable "STEM effect." Extrapolation to other national contexts should account for the baseline gender gap and the extent to which organic labor market signals have already begun closing it in the target discipline.

Single post-treatment cohort. The current design identifies the immediate enrollment response over a single post-treatment year. Whether this effect persists, amplifies through role models and peer network channels, or attenuates as the novelty of the incentive diminishes cannot be assessed with available data. Extending the observation window as subsequent graduation cohorts become available will be essential for evaluating long-run cost-effectiveness and informing decisions about

whether the policy should be continued, scaled, or redesigned for specific disciplinary contexts.

Section 6: Limitations

Several limitations of the present analysis merit explicit acknowledgment.

Data structure and enrollment imputation. The primary identification relies on graduation records from the MUR administrative database, as direct enrollment figures disaggregated by nationality and degree class are not publicly available at the required level of granularity. Enrollment year is accordingly imputed by subtracting the standard two-year duration from the observed graduation year. While this convention is standard in Italian higher education research and consistent with the structure of *Laurea Magistrale* programs, it introduces a degree of measurement error to the extent that a non-trivial share of students do not complete their degrees on schedule. If on-time completion rates differ systematically between STEM and non-STEM disciplines — a plausible concern given the reputational difficulty of engineering programs — the imputation could introduce differential timing noise. We cannot directly assess this possibility with the available data, though the high R^2 values across model specifications and the sharp post-2022 discontinuity are reassuring on this dimension.

Single post-treatment observation. Due to the two-year imputation lag and the 2024 data ceiling, our design observes only one post-treatment enrollment cohort (2022). This precludes any analysis of dynamic effects — whether the initial enrollment response persists, intensifies, or dissipates in subsequent years — and

makes it impossible to distinguish a permanent behavioral shift from a one-time anticipatory response to the newly introduced incentive. The policy implications of a sustained versus transitory effect differ substantially, and this question can only be resolved as subsequent graduation cohorts from 2025 and 2026 become available.

Contamination of the domestic placebo group. The domestic dataset (Laureati per Classe di Laurea) aggregates all students enrolled in Italian universities regardless of nationality. It therefore includes a small proportion of international students who, unlike the treatment group, may have received the enhanced STEM scholarship. Using the international graduation records as a benchmark, we estimate that international students account for approximately 4.0% of the domestic sample in 2021 and 4.3% in 2022, with a period average of 3.6% across 2013–2024. The inclusion of this subgroup technically biases the domestic DiD coefficient in a positive direction — attenuating, rather than inflating, the contrast between cohorts. Our null domestic results therefore represent a conservative lower bound on the absence of a domestic treatment effect, and the actual degree of null-ness among purely domestic students is if anything understated. This contamination does not threaten the validity of our main findings; it strengthens them.

Geographic and institutional heterogeneity. The STEM bonus amounts vary across regional DSU agencies, ranging from approximately €300 to €1,400 depending on ISEE bracket and region. Our analysis treats the policy as a uniform binary treatment, which abstracts from this subnational variation in transfer amounts. To the extent that higher bonus amounts generate larger enrollment responses — a testable dose-response prediction — our aggregate estimate reflects a weighted average effect across regions and funding tiers rather than the marginal effect of any

specific transfer level. Future work exploiting regional variation in bonus amounts could recover more precise estimates of the enrollment price elasticity.

Path dependency in Master's enrollment. A structural feature of our research design warrants emphasis: Master's programs exhibit substantial path dependency relative to Bachelor's programs. Entry into specialized Master's degrees — particularly in Engineering — typically requires a cognate Bachelor's degree, whereas fields such as management, economics, or social sciences admit students from a considerably broader range of undergraduate backgrounds. This asymmetry implies that the population of women eligible to respond to the STEM bonus at the Master's level is mechanically constrained by field choices made three to five years earlier, prior to the policy's existence. The 6.0 percentage point effect we identify therefore represents a lower bound on the potential enrollment response to financial incentives, as it captures only reallocation among women who had already committed to a compatible undergraduate pathway. A substantially larger effect might materialize at the Bachelor's level, where field choices are less constrained by prior academic investments and the pool of persuadable students is correspondingly wider. Testing this prediction constitutes a natural extension of the present work as Bachelor's-level graduation data for post-2022 cohorts become available (expected from 2025 onward, given the three-year program duration).

Section 7: Conclusion and Policy Implications

This study provides causal evidence that targeted financial incentives can meaningfully shift female enrollment in male-dominated STEM fields. Exploiting Italy's MUR Decree No. 1320 (2021) alongside a sharp discontinuity in scholarship

eligibility between international and domestic student populations, we document a 6.0 percentage point increase in the female share of Engineering enrollment among the highly treated international cohort — a finding robust to permutation-based inference, COVID-period sensitivity checks, and a domestic placebo test that rules out cultural confounding as an alternative explanation.

The policy's impact is not only statistically significant but also economically meaningful. Over the 2022–2024 period, the 6.0 percentage point increase in female Engineering enrollment translates into approximately 209 additional female Engineering graduates—a 17.5% gain relative to the counterfactual scenario without the scholarship. To put this in perspective, this means that for every 100 female Engineering graduates under the pre-policy trend, the scholarship generated an additional 17–18 graduates. These absolute gains highlight the potential of targeted financial incentives to meaningfully shift gender composition in structurally rigid fields, even in the short run.

The heterogeneity of treatment effects across disciplines carries the most actionable policy implication. Engineering responds strongly; Basic Sciences and ICT do not. This pattern suggests that the returns to financial incentives are not uniform across STEM but are instead concentrated in fields where structural barriers are high, gender gaps are persistent, and organic market signals have demonstrably failed to generate convergence. Applying a uniform scholarship bonus across all STEM disciplines — as the current Italian policy does — is therefore an inefficient allocation of public resources relative to a design that concentrates transfers in high-barrier fields. Policymakers seeking to maximize the enrollment impact of a given budget should treat discipline-level structural characteristics as a primary targeting criterion rather than an afterthought.

A related implication concerns the distinction between demand-side and supply-side constraints. The domestic ICT trend — where female enrollment began rising organically around 2018, likely driven by labor market expansion in digital technologies — illustrates that financial incentives are largely redundant in fields where market signals are already closing the gap. Public investment in STEM gender equity yields its highest marginal return where the market has failed to self-correct, not where it is already doing so.

Several important questions remain open. Whether the Engineering effect persists across subsequent cohorts, whether it amplifies through peer network and role model channels as female representation grows, or whether it attenuates over time cannot be determined from a single post-treatment observation. Equally, whether domestic Italian women would respond to an equivalent scholarship under full eligibility — the direct test of domestic price elasticity — requires either a means-testing reform or a regression discontinuity exploiting the ISEE threshold. These represent the natural next steps for this research agenda as additional graduation cohorts become available.

Ultimately, the evidence from Italy suggests that the gender gap in structurally rigid STEM fields is sensitive to price signals when those signals are sufficiently large and precisely targeted. Financial incentives are neither a universal solution nor an ineffective one — their impact depends critically on where they are applied.

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Appendix A: Data and Variable Definitions

A.1. Data Sources

The analysis relies on open-source administrative data provided by the Italian Ministry of University and Research (MUR).

- ***International Students (Treatment & Control):*** *Data on graduates (Laureati) from Master's degree programs (Laurea Magistrale), disaggregated by gender and degree class.*
    - *Source: USTAT Open Data Portal (dati.ustat.miur.it)*
- ***Domestic Students (Placebo):*** *Aggregated enrollment data (Immatricolati) for Italian students used for placebo tests.*
    - *Source: USTAT Open Data Portal (http://dati.ustat.miur.it)*

A.2. Classification Strategy

To ensure reproducibility, degree programs are classified into categories based on their Ministerial Degree Class codes (Classi di Laurea Magistrale - LM). The classification logic used in our analysis is detailed below:

Category	***Description***	***Ministerial Codes (LM)***
Engineering	*Core Engineering disciplines*	*LM-20 to LM-35 (inclusive)* *and LM-4, LM-53*
ICT	*Information & Comm. Tech.*	*LM-18, LM-66, LM-91*
Basic Sciences	*Basic Sciences and Mathematics*	*LM-17, LM-40, LM-44, LM-54, LM-58, LM-82, LM-6, LM-7, LM-8, LM-9, LM-60, LM-75*
NON_STEM_SOC	*Mixed*	*LM-16, LM-56, LM-76, LM-77, L-82, LM-83, LM-52, LM-62, LM-63, LM-81, LM-87, LM-88, LM-90, LM-1, LM-2, LM-5, LM-10, LM-11, LM-14, LM-15, LM-36, LM-37, LM-38, LM-39,*

		LM-43, LM-45, LM-64, LM-65, LM-80, LM-84, LM-89, LM-92, LM-93, LM-94, LM-50, LM-51, LM-57, LM-85

A.3. Policy Documentation

- ***MUR Decree No. 1320:*** *Official text of the decree establishing the STEM scholarship fund (Decreto Ministeriale n. 1320 del 17-12-2021).*

  Available at:

  https://www.mur.gov.it/it/atti-e-normativa/decreto-ministeriale-n-1320-del-17-12-2021